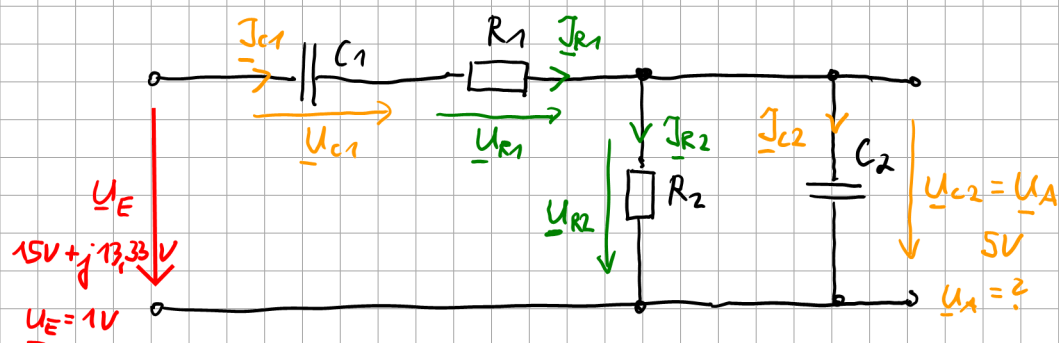


ZE 13



$$R_1 = R_2 = 3 \text{ k}\Omega$$

$$C_1 = C_2 = 22 \text{ nF}$$

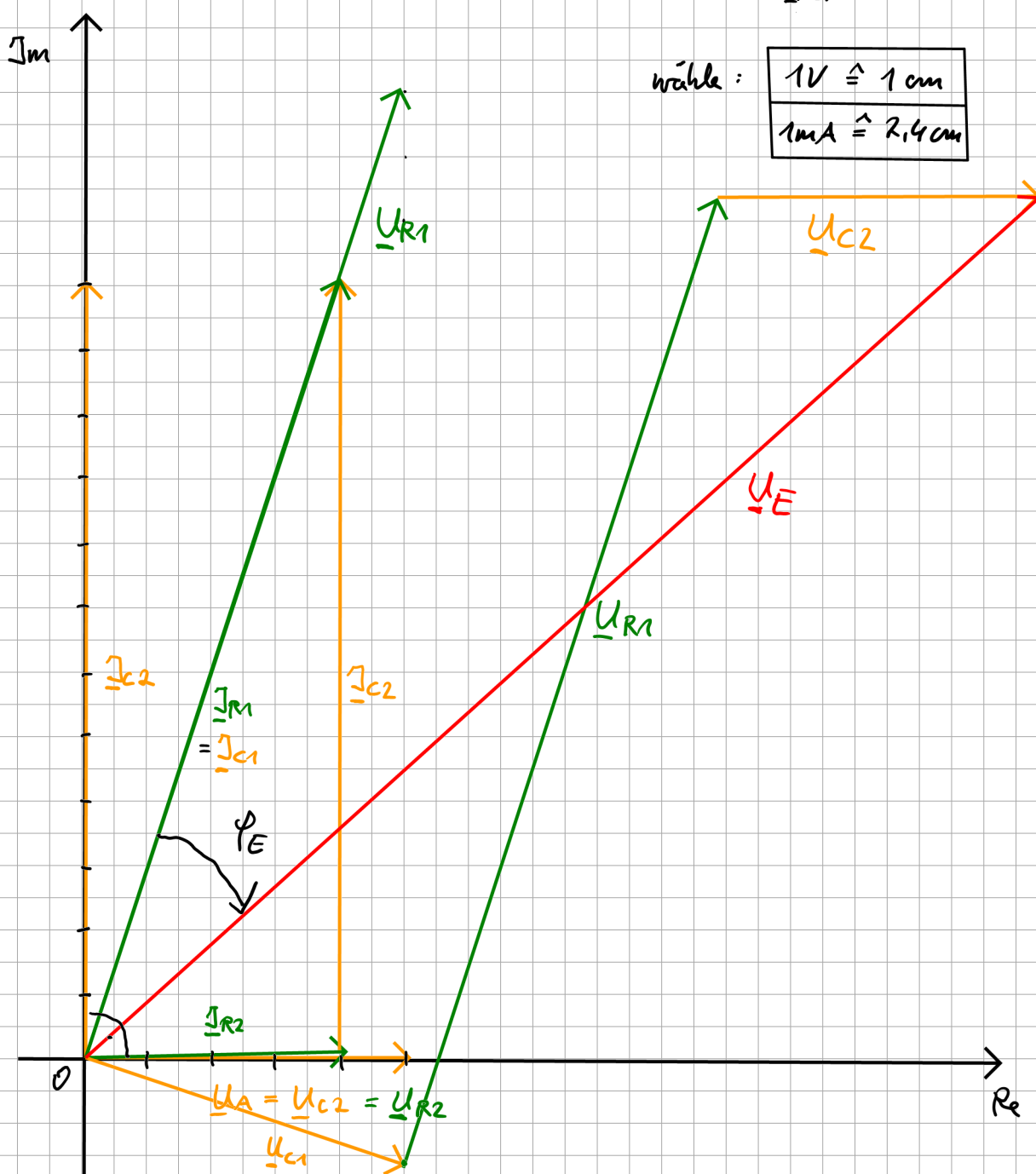
$$f = 7,24 \text{ kHz}$$

Vorgabe: $|\underline{U}_A| \hat{=} 5 \text{ cm}$, rechts

$$|\underline{I}_{R2}| \hat{=} 4 \text{ cm}$$

wähle:

$1V \hat{=} 1 \text{ cm}$
$1\text{mA} \hat{=} 2,4 \text{ cm}$



$$\underline{Z}_{C2} = \frac{1}{j\omega C_2} = \frac{1}{j2\pi f \cdot C_2} = \frac{1}{j2\pi 7,24 \text{ kHz} \cdot 22 \text{ nF}}$$

$$= \frac{1}{j} 1000 \cdot \frac{1}{\frac{1}{8} \frac{\text{As}}{\text{V}}}$$

$$\underline{Z}_{C2} = -j 1 \text{ k}\Omega$$

$$\underline{Z}_{C1} = \underline{Z}_{C2} = -j 1 \text{ k}\Omega$$

$$1 \text{ V} \hat{=} 1 \text{ cm} \Rightarrow \underline{U}_A = \underline{U}_{C2} = 5 \text{ V} + j0 \text{ V}$$

$$\underline{I}_{C2} = \frac{\underline{U}_{C2}}{\underline{Z}_{C2}} = \frac{5 \text{ V}}{-j 1 \text{ k}\Omega} = +j 5 \text{ mA} \quad (\hat{=} 12 \text{ cm})$$

$$\underline{U}_{R2} = \underline{U}_{C2} = 5 \text{ V}$$

$$\underline{I}_{R2} = \frac{\underline{U}_{R2}}{R_2} = \frac{5 \text{ V}}{3 \text{ k}\Omega} = \frac{5}{3} \text{ mA} = 1,67 \text{ mA} \quad (\hat{=} 4 \text{ cm})$$

$$\Rightarrow 1 \text{ mA} \hat{=} \frac{4 \text{ cm}}{\frac{5}{3}} = \frac{3}{5} \cdot 4 \text{ cm} = 2,4 \text{ cm}$$

$$\underline{I}_{R1} = \underline{I}_{R2} + \underline{I}_{C2}$$

$$= \underbrace{1,67 \text{ mA}}_{4 \text{ cm}} + \underbrace{j 5 \text{ mA}}_{12 \text{ cm}}$$

$$\underline{U}_{R1} = R_1 \cdot \underline{I}_{R1}$$

$$= 3 \text{ k}\Omega \cdot (1,67 \text{ mA} + j 5 \text{ mA})$$

$$= 5 \text{ V} + j 15 \text{ V}$$

$$\underline{U}_{C1} = \underline{Z}_{C1} \cdot \underline{I}_{C1}$$

$$= -j 1 \text{ k}\Omega \cdot (1,67 \text{ mA} + j 5 \text{ mA})$$

$$= -j 1,67 \text{ V} + 5 \text{ V}$$

$$\underline{U}_E = \underline{U}_{C1} + \underline{U}_{R1} + \underline{U}_A$$

$$= -j 1,67 \text{ V} + 5 \text{ V} + 5 \text{ V} + j 15 \text{ V} + 5 \text{ V}$$

$$= 15 \text{ V} + j 13,33 \text{ V}$$

$$b) \quad \underline{U}_A = 5V \quad \Rightarrow \quad \underline{U}_E = 15V + j13,33V$$

$$\underline{U}_A = x \quad \Leftarrow \quad \underline{U}_E = 1V$$

$$\frac{5V}{x} = \frac{15V + j13,33V}{1V}$$

$$\Rightarrow x = \frac{5V \cdot 1V}{15V + j13,33V}$$

$$= \frac{5V^2}{\sqrt{(15V)^2 + (13,33V)^2} \cdot e^{j \arctan(\frac{13,33}{15})}}$$

$$= \frac{5V^2}{20,1V \cdot e^{j41,6^\circ}}$$

$$= 0,25 \cdot e^{-j41,6^\circ}$$

$$c) \quad \varphi_E = \varphi_{\underline{U}_E} - \varphi_{\underline{I}_{c1}}$$

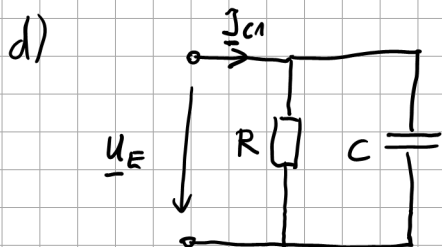
$$= \angle(\underline{U}_E) - \angle(\underline{I}_{c1})$$

$$= \angle(15V + j13,33V) - \angle(1,67mA + j5mA)$$

$$= \arctan\left(\frac{13,33}{15}\right) - \arctan\left(\frac{5}{1,67}\right)$$

$$= 41,6^\circ - 71,5^\circ$$

$$= -29,9^\circ$$



Parallelschaltung $\Rightarrow$ Admittanzen nutzen

$$\underline{Y}_E = \frac{\underline{I}_{c1}}{\underline{U}_E} = \frac{1}{R} + j\omega C$$

$$= \frac{1,67mA + j5mA}{15V + j13,33V} \stackrel{!}{=} \frac{1}{R} + j\omega C$$

$$= \frac{(1,67 \text{ mA} + j5 \text{ mA}) \cdot (15 \text{ V} - j13,33 \text{ V})}{(15 \text{ V} + j13,33 \text{ V})(15 \text{ V} - j13,33 \text{ V})} = \frac{1}{R} + j\omega C$$

$$= \frac{0,02505 \text{ A} \cdot \text{V} + j0,075 \text{ A} \cdot \text{V} - j0,02226 \text{ A} \cdot \text{V} + 0,0666 \text{ A} \cdot \text{V}}{(15 \text{ V})^2 + (13,33 \text{ V})^2}$$

$$= \frac{0,09165 \text{ A} \cdot \text{V} + j0,0527}{402,7 \text{ V}^2} = \frac{1}{R} + j\omega C$$

$$= 2,276 \cdot 10^{-4} \text{ S} + j1,309 \cdot 10^{-4} \text{ S} = \frac{1}{R} + j\omega C$$

$$\Rightarrow R = \frac{1}{2,276 \cdot 10^{-4} \text{ S}} = \underline{\underline{4,4 \text{ k}\Omega}}$$

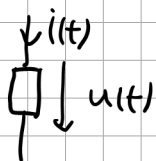
$$\omega C = 1,309 \cdot 10^{-4} \text{ S}$$

$$C = \frac{1,309 \cdot 10^{-4} \text{ S}}{2\pi \cdot 7,24 \text{ kHz}}$$

$$= \underline{\underline{2,9 \mu\text{F}}}$$

7 Leistung bei sinusförmiger Anregung

Allgemein für Leistung

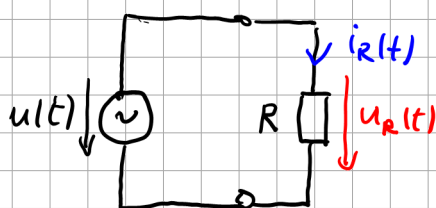


$$p(t) = u(t) \cdot i(t)$$

p
gilt immer

Augenblicksleistung
(instantaneous power)

Jetzt neu: $u(t)$ ist sinusförmig



$$u(t) = \hat{u} \cdot \sin(\omega t + \varphi)$$

$$= \sqrt{2} U \cdot \sin(\omega t + \varphi)$$

